Lesson notes · for you

Final project: let the computer find the best plan

Let the computer search: knapsack, routes, timetables: final project

Time
45 min

Materials
Your notebook and your work from the four lessons; A phone to take photos or record a voice note, if you have one

Learning goals

- The best trip bag under a weight limit
- The shortest route to visit three friends
- A club timetable with no clashes

Activity: Build it

1. Pick an idea and write the plan
2. Reuse your milestone work
3. Test it: where does it work, where does it fail?
4. Explain it to a partner in your own words

Discussion

- What would change if one part of your project changed?
- Where did it fail, and what did you change?
- What would you build next?

Check yourself

- One thing I understand now that I didn't before this project.

Try it this week
Explain your project to someone at home, and ask them one question it could answer.

Worksheet

Final project: let the computer find the best plan

Let the computer search: knapsack, routes, timetables: final project

Name _____ Class _____ Date _____

1 **Which shows best that you really understand your project?**

- ◯ It works when I run it
- ◯ I can explain how it works and what would happen if one thing changed
- ◯ It looks neat and professional
- ◯ An AI tool wrote a long explanation for it

2 **If you used an AI tool for part of your project, it's best not to mention it.**

- ◯ True
- ◯ False

3 **Which of these belong in your final project? Pick all that apply.**

- ☐ Work from your earlier milestones
- ☐ A test that shows where it works and where it fails
- ☐ Your own explanation of how it works
- ☐ A friend's project with your name on it